

MATH-150

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Turn your written portion in on a *separate* piece of paper.

Include a separate paper with your work. Show *all* your work in order to receive credit.

Save Yourself From Zombies!

There has been an outbreak of zombies at the University of San Diego. You, fortunately, have used your calculus skills to survive the outbreak. But you realize not everyone you know and love has the necessary calculus skills to survive.

This is what we know about zombies ... they have the following weaknesses [1]:

- They wander around in an awkward pattern that can be described by an equation.
 - They are not very clever. Thus, if you get into a fight with a zombie, even though they are strong, they will not be clever enough to pick up a weapon or guard themselves.
 - Zombies cannot swim.
 - Zombies are addicted to Snickers. This isn't a well known fact, but who isn't addicted to Snickers?
1. For your first encounter, you realize that the zombies are wandering around in an awkward pattern described by the following function:

$$(x^2 + y^2 - 1)^3 = -x^2y^3$$

They do not move very efficiently. Anyone that stays tangent to this path will be safe and is able to evade a single zombie attack.

- (a) You have teamed up with 5 friends to survive the apocalypse. Who are the 5 people on your team?
- (b) Create a plan for how you and your friends should evade this zombie. In order for all your friends to survive, you must write up your plan and turn it in to your professor by Friday, November 4. You may turn your plan in sooner to receive feedback and get details about your next zombie encounter.

Future Zombie Encounters

1. Walking a tangential path. Due November 4.
3. Trapping the zombies in water. Due November 11.
2. Using a weapon against the zombies. Due November 11.
4. The quickest path to safety. Due November 15.

The final memoir is due Tuesday, November 22.

References

- [1] Offutt, J, "How to Kill Monsters Using Common Household Items" 2015, Permuted Press

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 - They are not very clever. Thus, if you get into a fight with a zombie, even though they are strong, they will not be clever enough to pick up a weapon or guard themselves.
 - Zombies cannot swim.
 - Zombies are really addicted to Snickers. This isn't a well known fact, but who isn't addicted to Snickers?
2. Hiding and fleeing zombies has created quite a hunger for you and your team. As you walk towards the UC in search for food, you stumble upon a strange wall of zombies standing 10 feet in front of you with their backs towards you. The wall can be described with the function $y^3 = x^2$. You know the only way to survive and get to the UC is to go through this wall of zombies. Yet, you are very tired so you need to find the zombies closest to you and attack there.
- (a) Are any of your teammates zombies? Who is left in the group?
- (b) Create a plan for how you and your friends should attack these zombies. In order for all your friends to survive, you must write up your plan and turn it in to your professor by Friday, November 11. You may turn your plan in sooner to receive feedback and get details about your next zombie encounter.

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1. Walking a tangential path. Due November 4.
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3. During your next encounter, you are chased into the pool area at the Sports Center by a group of zombies. Some survivors have drained and reconfigured the pool so that is now shaped like an inverted cone with radius 41 feet and 9 feet deep. You and your team realize that you can trap the zombies in the pool. If you and your friends can fill the pool at a rate of 1000 ft^3 per minute, then you can calculate the rate the height of the water is changing when the tallest zombie is trapped.

- (a) Are any of your teammates zombies? Who is left in the group?
- (b) Create a plan for how you and your friends should trap these zombies. In order for all your friends to survive, you must write up your plan and turn it in to your professor by [encounter 3 due date]. You may turn your plan in sooner to receive feedback and get details about your next zombie encounter.

Future Zombie Encounters

1. Walking a tangential path. Due [encounter 1 due date].
2. Using a weapon against the zombies. Due [encounter 2 due date].
3. Trapping the zombies in water. Due [encounter 3 due date].
4. The quickest path to safety. Due [encounter 4 due date].

The final memoir is due [memoir due date].

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4. After weeks of evading zombies, you and your team have collected enough intel to realize the zombies tend to wander towards lower altitudes. You've heard the mantra, "when they go low, we go high" [2].

You can view any point on campus as being in the domain (input) of a function A whose range (output) is the amount of meters that point is above (or below) sea level. Assume the campus boundaries are those shown in the map in Figure 1. Is there a point on campus where the zombies will eventually congregate? Explain. Is there a safest place on campus where all survivors should meet? Explain.

- (a) Are any of your teammates zombies? Who is left in the group?
- (b) Create a plan for how you and your friends will find safety. In order for all your friends to survive, you must write up your plan and turn it in to your professor by Wednesday, November 16. You may turn your plan in sooner to receive feedback.

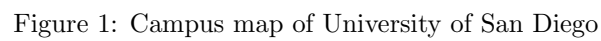
Future Zombie Encounters

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References

- [1] Offutt, J, "How to Kill Monsters Using Common Household Items" 2015, Permuted Press
- [2] Obama, Michelle, "Democractic National Convention 2016 Speech." July 25, 2016.

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Each of the parts of the project have been graded by your professor at least once. Consider any feedback from previous submission as you compile your encounters into a full memoir. Someone reading your memoir in the future should be able to follow your methods in order to survive a zombie outbreak. The final memoir will be evaluated according to the attached rubric. For your final project, keep the following in mind:

1. Your computations should be written in a clear and organized manner where each step is justified or clearly follows from previous steps.
2. Your plan should provide reasoning for any information used in the computation.
3. You must address any feedback given on your previous submissions.
4. Your memoir should be a cohesive story. Don't forget to say how the story ends!
5. Include all versions of all 4 parts. Include scratch work.

References

- [1] Offutt, J, "How to Kill Monsters Using Common Household Items" 2015, Permuted Press